

Analysis of Algorithms (BDSF25M/A)

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Quiz 03

Time: 15 min

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Grading	CLO-1	CLO-2	CLO-3	CLO-4
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Ternary Heaps are like binary heaps, in which every node can have three children left, mid and right. Rest the concepts, terms and rules are similar/extended from binary heaps. You need to describe the formula for left, mid, right child and parent nodes, also altered heapify and insert functions for the ternary heaps.

Formulas:

$$\text{parent}(i) = \lfloor (i+1)/3 \rfloor$$
$$\text{left}(i) = 3i - 1$$
$$\text{right}(i) = 3i + 1$$
$$\text{mid}(i) = 3i$$

Heapify:

max-heapify-ternary(A, i)

- 1 $l = \text{left}(i)$
- 2 $m = \text{mid}(i)$
- 3 $r = \text{right}(i)$
- 4 $\text{largest} = i$
- 5 if $l \leq \text{heap-size}[A]$ and $A[l] > A[\text{largest}]$
- 6 $\text{largest} = l$
- 7 if $m \leq \text{heap-size}[A]$ and $A[m] > A[\text{largest}]$
- 8 $\text{largest} = m$
- 9 if $r \leq \text{heap-size}[A]$ and $A[r] > A[\text{largest}]$
- 10 $\text{largest} = r$
- 11 if $\text{largest} \neq i$
- 12 Swap $A[i]$ with $A[\text{largest}]$
- 13 max-heap-ternary(A, largest)

Insert :

ternary - max - heap - insert (A, key)

- 1) $\text{heap-size}[A] = \text{heap-size}[A] + 1$
- 2 $i = \text{heap-size}[A]$
- 3 $A[i] = -\infty$
- 4 while $i > 1$ and $A[\text{parent}(i)] < \text{key}$
- 5 $A[i] = A[\text{parent}(i)]$
- 6 $i = \text{parent}(i)$
- 7 $A[i] = \text{key}$

Time Complexity: For ternary heap

Heapify : $O(\log_3 n)$

Insert : $O(\log_3 n)$